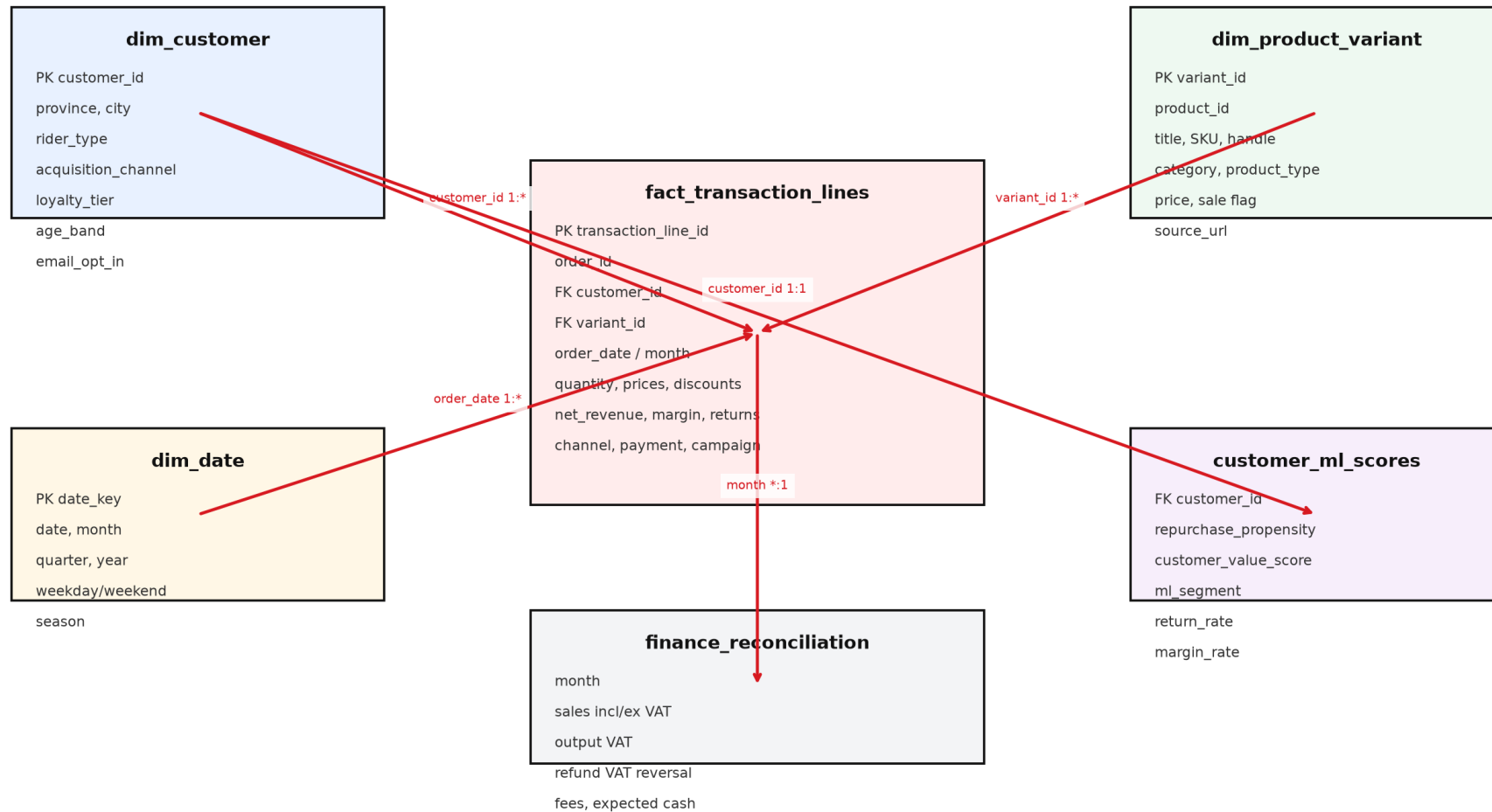


Leatt BI/ML ERD and Fabric Deployment Plan

Star schema ERD, million-row data proof and Fabric Data Factory / Lakehouse deployment path.

Leatt Ecommerce BI/ML Star Schema ERD

Million-row ecommerce fact table with customer, product, date, ML and finance reconciliation marts



Tables

name	type	grain	rows	keys	description	
fact_transaction_lines	Fact	One ecommerce order line	2,000,000	transaction_line_id PK; order_id; customer_id FK, order_id, discount, revenue, margin, return, payment and campaign fact.		
dim_customer	Dimension	One synthetic customer	180,000	customer_id PK	Customer geography, rider type, acquisition channel, loyalty tier, age band and opt-in status.	
dim_product_variant	Dimension	One Shopify product variant	11,354	variant_id PK; product_id	Extracted Leatt product/variant catalog with category, SKU, price, availability and source.	
dim_date	Dimension	One calendar date	912	date_key PK	Calendar/date attributes derived from order_date for time intelligence.	
customer_ml_scores	ML Output	One customer score	180,000	customer_id FK	Repurchase propensity, value score, margin rate, return rate and ML segment.	
agg_monthly/category/product/province/country	Table Aggregates	Reporting aggregate	Small	various	Power BI-ready aggregate tables for executive dashboards.	
finance_reconciliation	Finance Mart	One accounting period	30	month	VAT, refunds, sales excl/incl VAT, payment fees and expected cash reconciliation.	

Relationships and Fabric Status

from_table	from_column	to_table	to_column	cardinality
dim_customer	customer_id	fact_transaction_lines	customer_id	1:*
dim_product_variant	variant_id	fact_transaction_lines	variant_id	1:*
dim_date	date_key	fact_transaction_lines	order_date	1:*
dim_customer	customer_id	customer_ml_scores	customer_id	1:1
fact_transaction_lines	month	finance_reconciliation	month	*:1

Azure CLI is signed in and Fabric REST API can list workspaces.

Workspaces found: My workspace and Apollis.

Lakehouse creation returned FeatureNotAvailable in both workspaces, indicating Fabric capacity/feature enablement is required before pushing the million-row parquet into OneLake.

Once capacity is enabled, upload the 2M-row parquet into OneLake Bronze and load it to a Delta table using Fabric Data Factory or Lakehouse Load Table.